

- 9 (a) (i) stress = F / A C1
 = $25 / (1.7 \times 10^{-6})$
 = $1.47 \times 10^7 \text{ Pa}$ (do not allow 1 sig fig) A1
- (ii) stress = $E \times \text{strain}$ C1
 $1.47 \times 10^7 = 7.1 \times 10^{10} \times (\Delta l / 1.8)$
 $\Delta l = 0.37 \text{ mm}$ A1 [4]
- (b) $R = \rho l / A$ OR $R \propto L$ C1
 so, $\Delta R / R = \Delta l / l$ C1
 $\Delta R = (3.7 \times 10^{-4} / 1.8) \times 0.03 = 6.2 \times 10^{-6} \Omega$ A1 [3]

May calculate $\rho = 2.833... \times 10^{-8} \Omega \text{ m}$
 giving new R as $3.0006167 \times 10^{-2} \Omega$
 hence ΔR - full credit possible

However, if rounds off ρ as $2.83 \times 10^{-8} \Omega \text{ m}$,
 then $R_{\text{new}} < R_{\text{old}}$!
 Allow 1 mark only for $R \propto L$